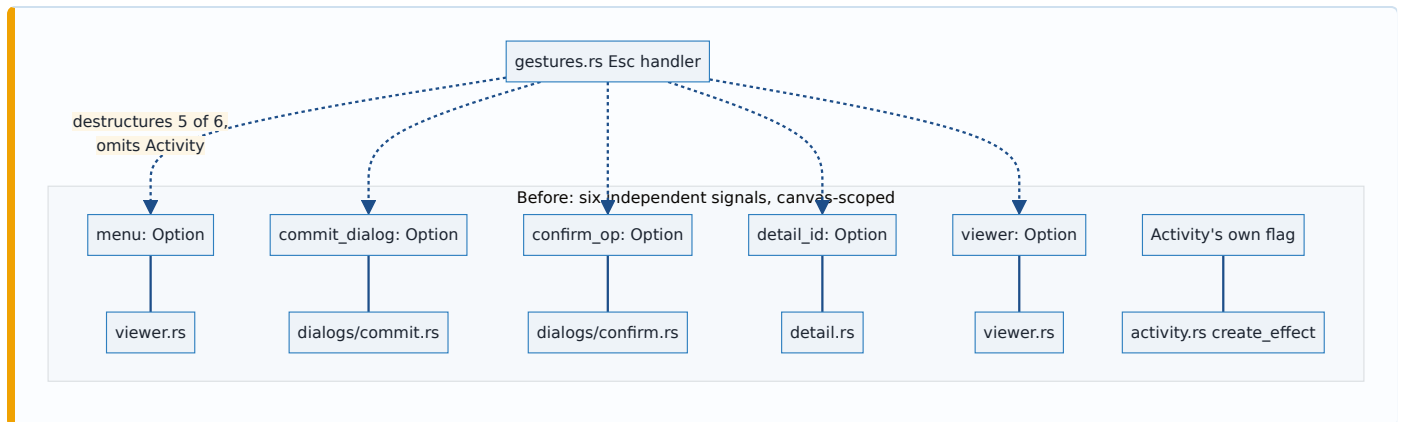


0024 — Frontend overlay state moves into a Dock -keyed OverlayStack

- **Status:** Accepted
- **Date:** 2026-07-27
- **Milestone / issue:** M1.11 — Refactor frontend state into feature boundaries (#64)
- **Supersedes:** nothing. **Amends:** the ad-hoc `Overlays` bundle in `state.rs` that held six independent overlay signals with no rule tying them together.
- **Related:** [0012](#) (the app shell), ADR 0005 / 0014 (LAN-view read-only affordances — see "Where this is implemented" for the session-side half of this refactor).

Context

Before M1.11, "what's currently covering the canvas" had no single owner. Six independent signals played that role — `Overlays::menu`, `commit_dialog`, `confirm_op`, `detail_id`, `viewer`, and `Activity`'s own open flag — each opened and closed by whichever call site happened to touch it, all born inside canvas scope so an epoch bump destroyed and recreated them.



Two bugs fell straight out of that shape, and neither was a one-off mistake — both were structural:

1. **Esc could not close the Activity panel.** `gestures.rs`'s Esc handler destructured `menu`, `commit_dialog`, `confirm_op`, `detail_id`, `viewer` in a hand-written five-way `if/else if` chain. `Activity` was simply the field nobody remembered to add — there was no list of overlays to be incomplete against.
2. **Two right-edge panels could render together for one frame.** The `Activity` panel and the `commit detail` panel both dock the right edge of the screen. `Overlays::open_detail_panel` closed `Activity` *synchronously* when the detail panel opened. The reverse direction ran from a `create_effect` in `activity.rs` that fired one reactive tick *after* `Activity`'s visibility flipped — so

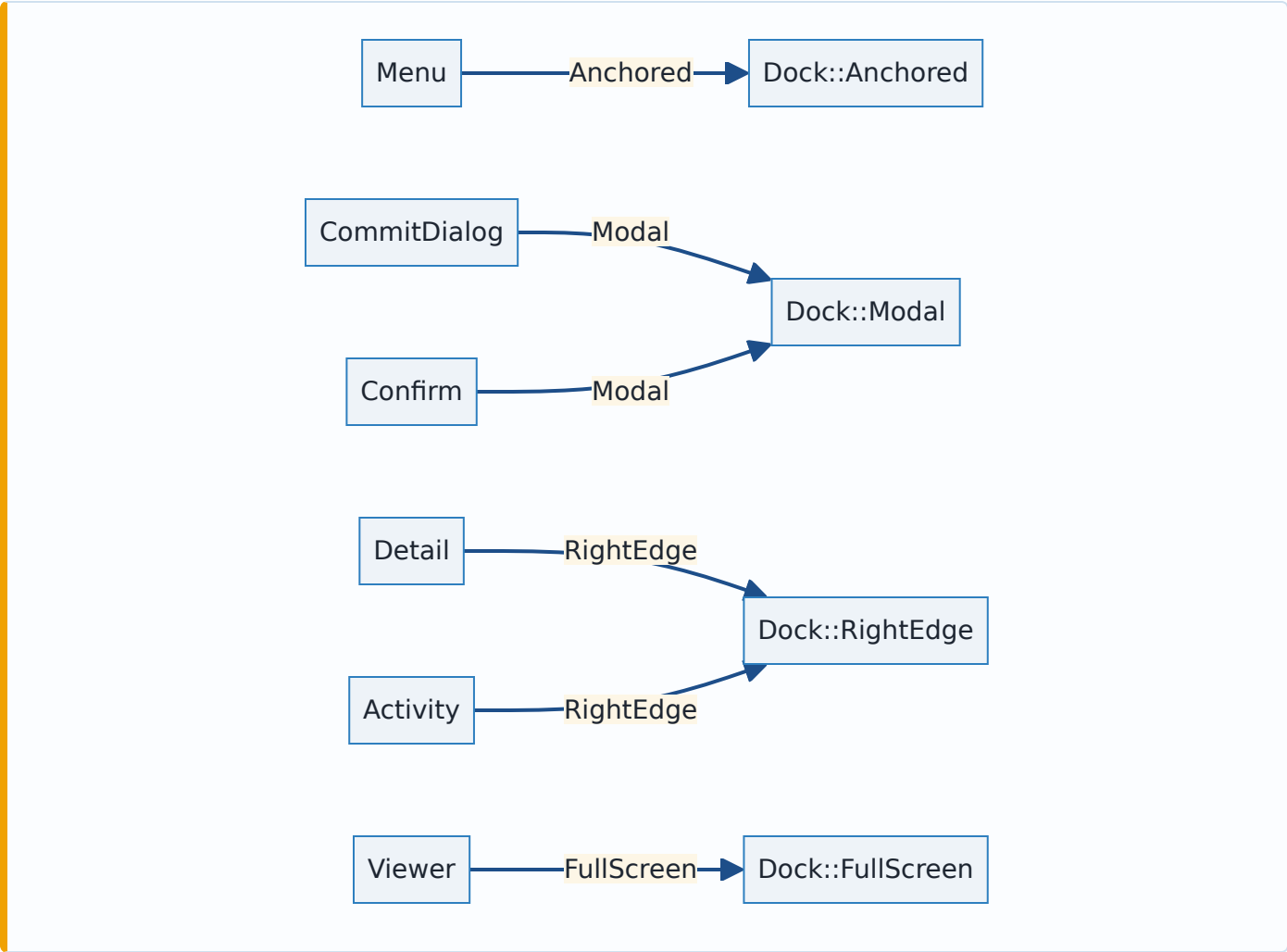
opening Activity while the detail panel was open left both visible for a frame, and the asymmetry between "synchronous" and "next tick" was exactly the kind of fact that stops being true the next time either call site is touched.

Decision

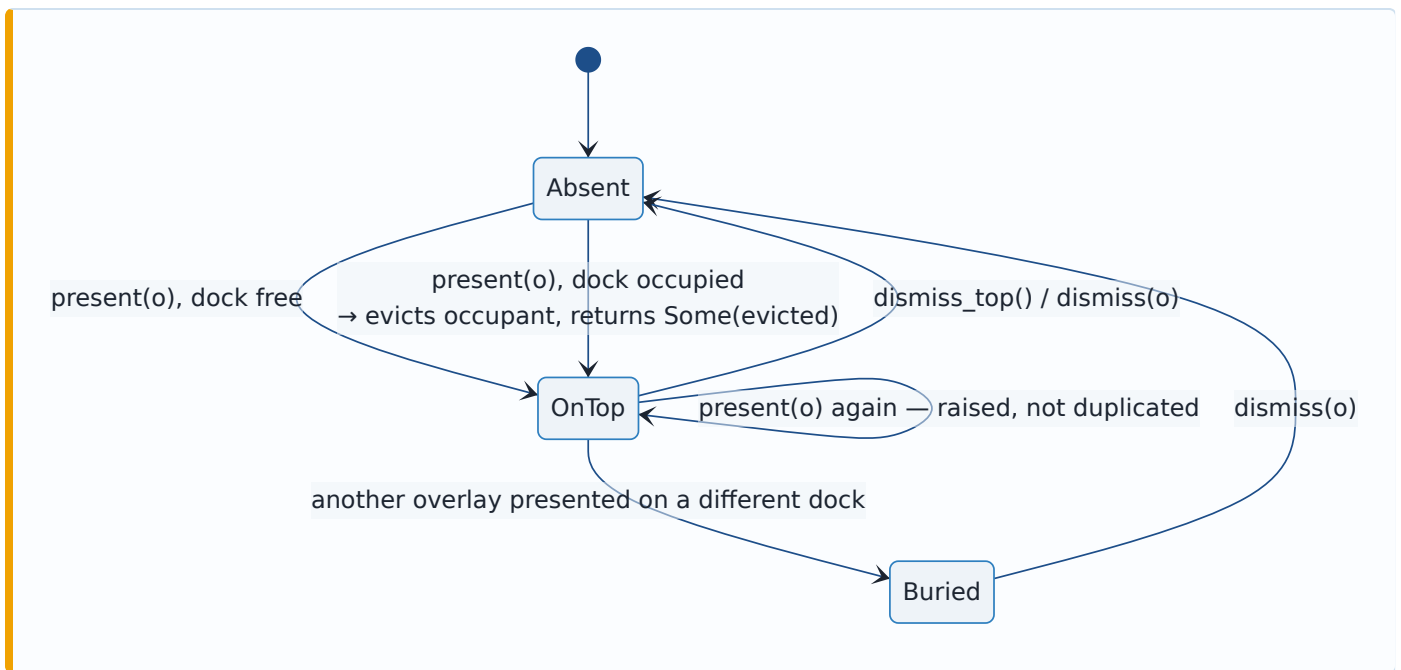
Give every overlay a single stack, keyed by **where it docks**, and make "at most one overlay per dock" a type-level invariant rather than a habit maintained by hand at each call site.

Overlay and Dock — framework-free, host-tested

Overlay (Menu , CommitDialog , Confirm , Detail , Viewer , Activity) maps to one of four Dock s (RightEdge , Modal , Anchored , FullScreen). Two overlays that resolve to the same dock cannot coexist; two on different docks can — which is exactly how the context menu (Anchored) legitimately floats over a right-docked panel today.



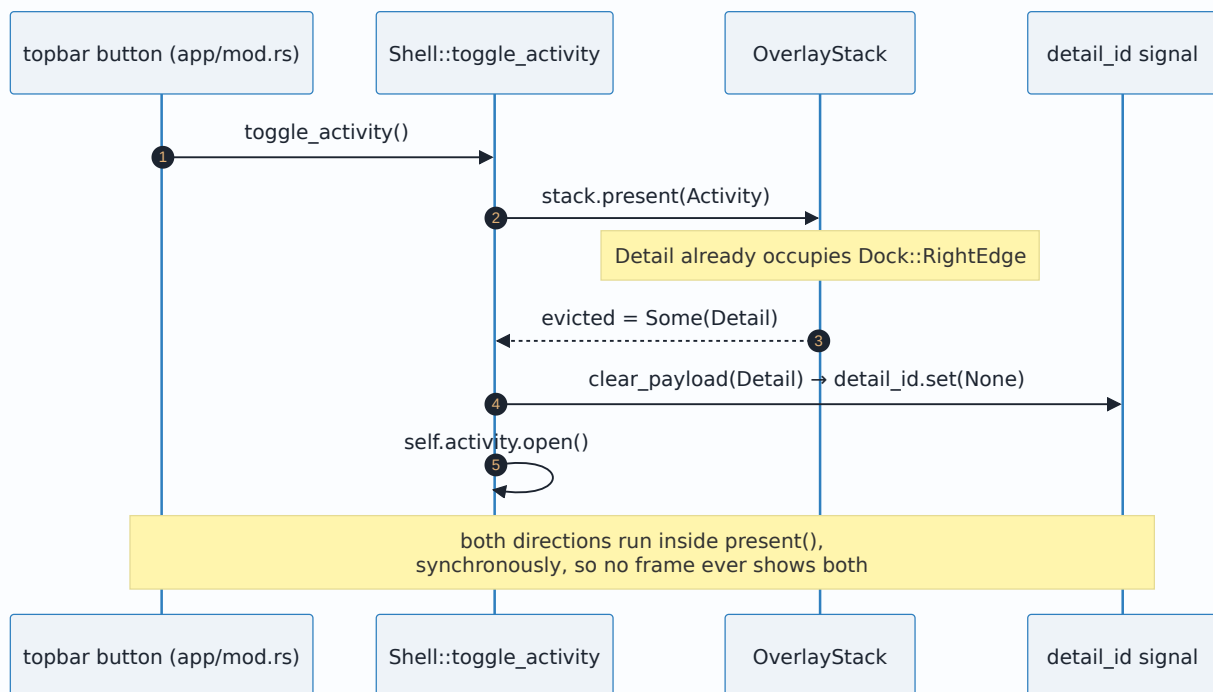
OverlayStack::present is the *only* way to add an overlay:



Dismissal is LIFO (`dismiss_top` pops the last element), which reproduces the priority the old hand-written Esc chain spelled out — viewer, then menu, then modals, then detail panel — not by copying that order as a constant but because it is exactly the order the real open sequences produce: the viewer only ever opens *from* an already-open detail panel, a modal only ever opens over whatever else is up. The one thing that *changes* rather than reproduces: Activity is now reachable by Esc at all.

Shell — the one place the stack and the payloads move together

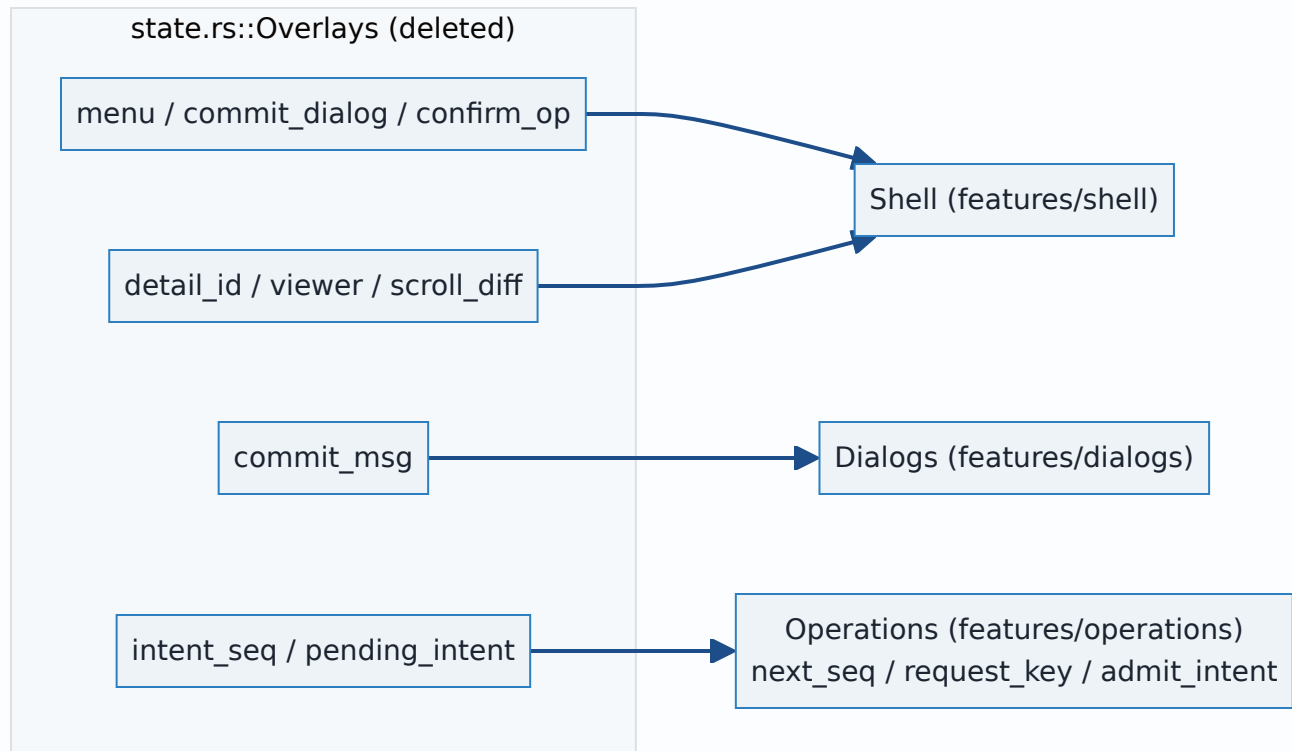
`features/shell/signals.rs`'s `Shell` holds the stack signal plus the six overlay payload signals (`menu`, `commit_dialog`, `confirm_op`, `detail_id`, `viewer_doc`, and a handle to `Activity`), **all private**. The public entry points that touch the stack are `open_*` and `toggle_activity` (which call `present`), `close_*` (which call `dismiss`), `dismiss_top`, and `reads`. `present`, `dismiss` and `dismiss_top` are the only three functions that call `clear_payload`, so a payload can never go stale relative to what the stack says is showing.



That sequence is the fix for bug 2 above: before M1.11, "Activity evicts Detail" ran a tick late because it lived in a `create_effect` ; now both directions of right-edge eviction are the same code path, `present` , so there is no asymmetry left to drift.

Where the six overlay signals now live

`Overlays` is gone from `state.rs` . Its thirteen fields are re-homed:



Features gains shell and loses activity — it is one of the six overlays now, though App still holds the Activity handle directly too, because the shared status read keys on it.

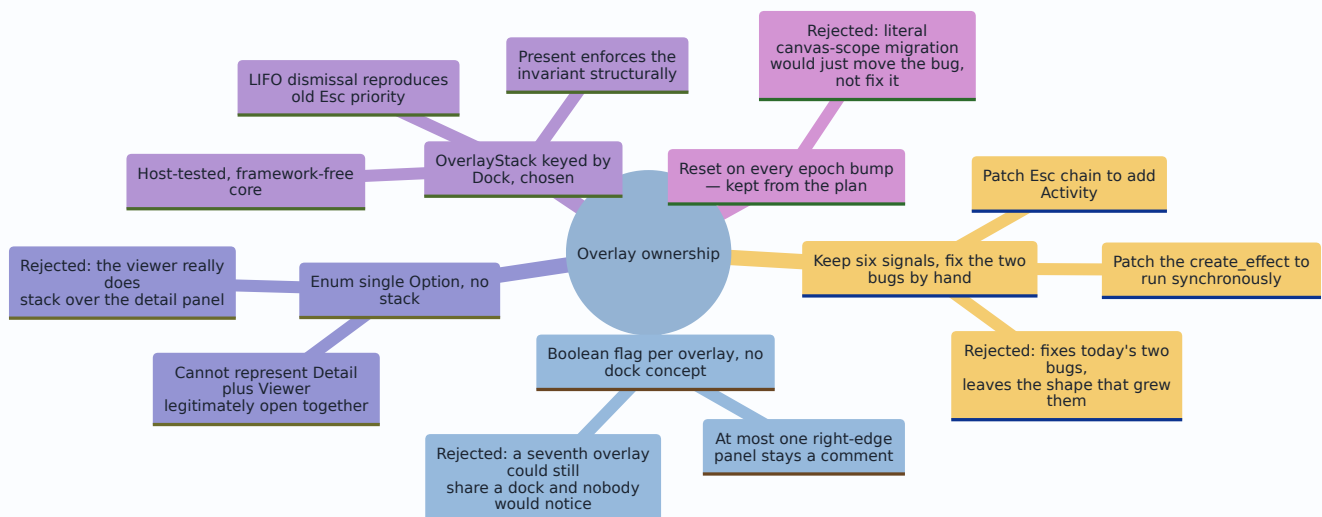
Consequences

1. **Esc closes the Activity panel.** Proven by a host-side test (`escape_dismisses_the_activity_panel`), not just fixed by inspection.
2. **The two right-edge panels can never both be visible.** Both eviction directions run through the same synchronous `present` ; the frame where both used to render is gone by construction, not by timing luck.
3. **Overlays now survive an epoch bump — a deliberate behavior change.** Shell is created once in App , above graph_canvas , for the same reason Features is. Before M1.11 the detail panel, viewer, menu and modals lived inside canvas scope and were destroyed on every epoch rebuild; now they are not. The detail panel's Resource re-keys and refetches on its own, so it self-heals or shows an honest error if the commit is gone across the rebuild. Same reasoning covers the commit draft (`commit_msg`) and the click-ordering pair: the old canvas-scope comment argued reset-on-rebuild was correct, but not resetting is equally correct here, because an intent from a previous epoch already fails `RequestKey::is_current` and sequences only increase — nothing downstream depends on the reset actually happening.
4. **present returns Option<Overlay> , not ()** . This was a deliberate API choice, not an oversight: a () return would let eviction happen silently, and the caller — Shell , in this case — must know exactly which overlay was evicted so it can blank that overlay's payload signal. A silent

eviction would leak a stale payload (e.g. a detail panel's commit id lingering in `detail_id` after Activity evicted it from the screen).

5. **Dock is new vocabulary the original plan did not name.** It is what turns "two overlays that occupy the same place cannot both be visible" into a type the compiler can check, instead of a rule stated only in a comment at each of four call sites.
6. **This is a client-side, host-testable refactor.** `core.rs` has no `wasm32` dependency and its eleven tests (including `at_most_one_overlay_per_dock_after_any_sequence`, which drives a mixed sequence of presents and dismisses and asserts the invariant on the resulting end state) run natively. `signals.rs` is `#[cfg(target_arch = "wasm32")]` only — the invariant is proven off-target, the wiring is wasm-only glue.
7. **A parallel, smaller instance of the same idea landed in `features/session`.** The frontend's CSRF token and `via_lan` flag used to be two independent `thread_local!`s in `api.rs` with no rule tying them together. `SessionCore` now holds both and adds a typed rejection, `SessionRejection::UiModeChangeWhileLan`, so "a LAN-view session may not select Active mode" is answerable from one place instead of an ad-hoc check. See "Where this is implemented" and the `SECURITY_MODEL.md` annotation below — this reinforces, client-side, the write-boundary ADR 0005/0014 already establish server-side; it does not change that boundary, which remains the LAN router's absent write routes.

Alternatives considered



Alternative	Why not
Patch the two bugs in place, keep <code>Overlays</code>	Fixes today's symptoms; leaves the shape — thirteen ungoverned fields — that produced them and will produce the next one.
A boolean/ <code>Option</code> per overlay with no <code>Dock</code> concept	"At most one right-edge panel" stays a fact enforced by memory at each call site, exactly the failure mode this ADR exists to remove.
Reset overlay state on every epoch bump, as the plan's literal Step migration implied	Rejected as <i>the</i> fix for Task 6's deferred step — moving the six signals out of canvas scope necessarily changes when they die. The resource re-key argument (Consequence 3) makes not-resetting sound rather than merely convenient.
<code>present</code> returns <code>()</code>	Silent eviction — the exact shape of bug this ADR fixes for the <i>stack</i> , reintroduced one layer down for <i>payloads</i> . Rejected; see Consequence 4.

Deviations from the plan, accepted

1. **The plan's Step 1 test 2 could not be built as written.** It presents `Activity` then `Detail` and asserts `Activity` survives underneath. Under the eviction rule those two can never coexist — that *is* the fix. The test keeps its intent (LIFO dismissal) with a genuinely stacking pair, `Detail` then `Viewer` (the viewer really is opened *from* the detail panel). Test 3 likewise uses `Detail / Menu` — different docks, so nothing is evicted and the test is purely about raise-not-duplicate.
2. **`present` returns `Option<Overlay>`, not `()`.** See Consequence 4.
3. **Behavior change, deliberate: overlays now survive an epoch bump.** See Consequence 3. Before M1.11 the detail panel, viewer, menu and modals were destroyed on every canvas rebuild; now they are not, because moving the signals out of canvas scope has no other honest reading.
4. **`Dock` is new vocabulary the plan did not name.** It is what makes "two overlays that occupy the same place cannot both be visible" a type-level fact rather than a habit — see Consequence 5.

Where this is implemented

Concern	Path
Overlay , Dock , OverlayStack , eviction rule, 11 host tests	crates/git-vista/src/features/shell/core.rs
Shell — private payload signals, present / dismiss / clear_payload	crates/git-vista/src/features/shell/signals.rs (# [cfg(target_arch = "wasm32")])
module wiring	crates/git-vista/src/features/shell/mod.rs
Features gains shell , loses activity as a direct field	crates/git-vista/src/state.rs
Esc → shell.dismiss_top() , replacing the five-way if/else if chain	crates/git-vista/src/gestures.rs
Shell created once, above graph_canvas	crates/git-vista/src/app/mod.rs , crates/git-vista/src/app/canvas.rs
call-site rewires (menu.set(None) → shell.close_menu() , 19 sites, etc.)	crates/git-vista/src/menu.rs , detail.rs , viewer.rs , activity.rs , dialogs/commit.rs , dialogs/confirm.rs , render/nodes.rs , render/stubs.rs
commit_msg re-homed	crates/git-vista/src/features/dialogs/{core,signals}.rs
intent_seq / pending_intent re-homed as next_seq / request_key / admit_intent	crates/git-vista/src/features/operations/{core,signals}.rs
the parallel SessionCore (CSRF token + via_lan , SessionRejection::UiModeChangeWhileLan)	crates/git-vista/src/features/session/{core,signals}.rs , wired from crates/git-vista/src/session.rs , crates/git-vista/src/api.rs , crates/git-vista/src/picker.rs , crates/git-vista/src/app/mod.rs
live evidence	docs/superpowers/evidence/2026-07-27-m1.11-live-drive.md (behaviours a/c/d driven live; b by code reading + host test)

Signed: thomas2025 · 2026-07-27T09:00:00-04:00